

Name: _____ (please print)

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Signature: _____

Date: _____

ECE 3364 – Quiz #4

Spring 2026

Do not open this quiz until you are told to begin.

1. Print your name, and sign your name, at the top of this page.
2. This quiz is closed book, closed notes. You may use one $8.5'' \times 11''$ crib sheet, or its equivalent. You may use a calculator. You should not use a cell phone, tablet computer, or laptop computer as you work on this quiz.
3. Show all work on these pages, and do NOT use backside of each page. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit. You may separate the pages as you work.
4. Show all units in solutions, intermediate results, and figures. Units in the quiz will be included between square brackets.
5. Do not use red ink. Do not use red pencil.
6. You will have 30 minutes to work on this quiz.

_____/100

Room for extra work

The circuit shown in Figure 1 operates in sinusoidal steady state. The signal source is

$$v_s(t) = 30 \cos\left(500 \frac{\text{rad}}{\text{s}} t\right) [\text{V}],$$

with internal resistance $R_s = 2 \, [\Omega]$. The circuit contains two ideal transformers with turns ratios 1:2 and 1:5 as shown. The middle branch consists of a $10 \, [\Omega]$ resistor in series with a $C = 500 \, [\mu\text{F}]$ capacitor. The load consists of a $550 \, [\Omega]$ resistor in series with an $L = 400 \, [\text{mH}]$ inductor. The current $i_B(t)$ is the current flowing through the middle branch, with the reference direction shown.

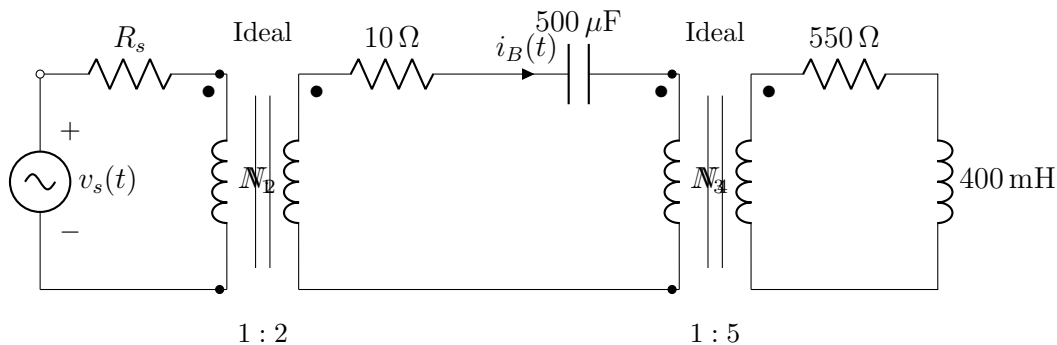


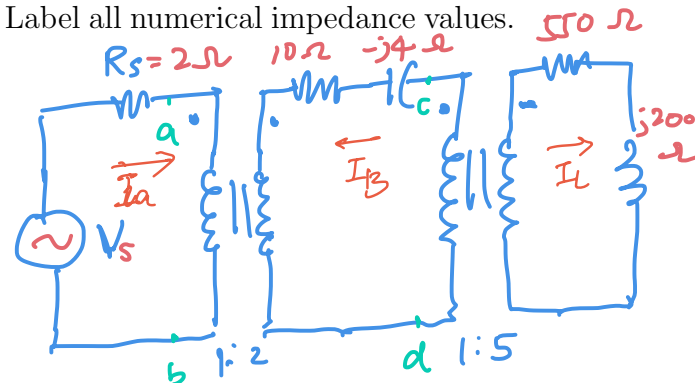
Figure 1

- a) (20 pts) Redraw the circuit in the phasor domain. Label all numerical impedance values.

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 500 \cdot 500 \times 10^{-6}} = \frac{1}{j0.25} = -j4 \, \Omega$$

$$Z_L = j\omega L = j(500)(0.4) = j200 \, \Omega$$

$$V_s = 30 \angle 0^\circ. \quad \text{Note, this is max value. Not rms}$$



- b) (20 pts) Find the numerical value of the impedance Z_{in} seen by the signal source. Express your answer in both rectangular and polar form.

Let Z_{cd} be the impedance reflected through the 1:5 ideal transformer

$$Z_{cd} = \frac{550 \, \Omega + j200 \, \Omega}{5^2} = 22 + j8 \, \Omega$$

Now add the middle branch in series:

$$Z_{mid} = (10 - j4) + (22 + j8) = 32 + j4 \, \Omega$$

now reflected through the 1:2 transformer:

$$Z_{ab} = \frac{Z_{mid}}{2^2} = \frac{32 + j4}{4} = 8 + j \, \Omega \Rightarrow \text{polar format}$$

If you consider R_s as well seeing by the voltage source

$$\text{You will get } Z_{in} = Z_{ab} + Z_s = 10 + j \, \Omega$$

$$\approx 10.05 \angle 5.71^\circ \, \Omega$$

- c) (20 pts) Find the phasor current $\mathbf{I}_a$ supplied by the signal source, and the phasor current $\mathbf{I}_B$ in the middle branch. Express each in polar form.

$$\mathbf{I}_a = \frac{\mathbf{V}_s}{R_s + Z_{ab}} = \frac{30 \angle 0^\circ}{10 + j1} = \frac{30(10 - j)}{101} = 2.97 - j0.29 \text{ A}$$

In polar format.

$$\mathbf{I}_a = \frac{30 \angle 0^\circ}{\sqrt{101} \angle 5.71^\circ} = 2.985 \angle -5.71^\circ \text{ A}$$

middle branch,

Ideal

$$I_1 N_1 = I_2 N_2$$

Note: dot convention, current enter in the dot is primary, leave at un-dot is secondary

$$\therefore \mathbf{I}_B = \frac{\mathbf{I}_a}{2} = 1.493 \angle -5.71^\circ \text{ A}$$

- d) (20 pts) Write the time-domain expression for $i_B(t)$.

Convert $\mathbf{I}_B$ to time domain.

magnitude 1.493.

phase -5.71°

$\omega = 500 \text{ rad/s}$

$$i_B(t) = 1.493 \cos(500t - 5.71^\circ) \text{ A}$$

e) (20 pts) Find the average power delivered to the $550\ [\Omega]$ load resistor.

Average power., we need the current in the load. I_L

$$P_{avg} = \frac{1}{2} |I_L|^2 \cdot R_L$$

$R_L = 550\ \Omega$, and we need to assign a reference sign for current I_L . But since taking magnitude, so the sign is NOT really matter. in this case.

using the Ideal transform of 1:5

$$I_L = \frac{I_B}{5} = \frac{1.493 \angle -51.71^\circ}{5} = 0.299 \angle -51.71^\circ\text{ A}$$

$$\therefore P_{avg} = \frac{1}{2} |I_L|^2 \cdot R$$

$$= \frac{1}{2} (0.299)^2 \cdot 550$$

$$\doteq 24.52\text{ W}$$

Note, if you use RMS value, then

$$P_{avg} = |I_L|_{rms}^2 R$$

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Page 6

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